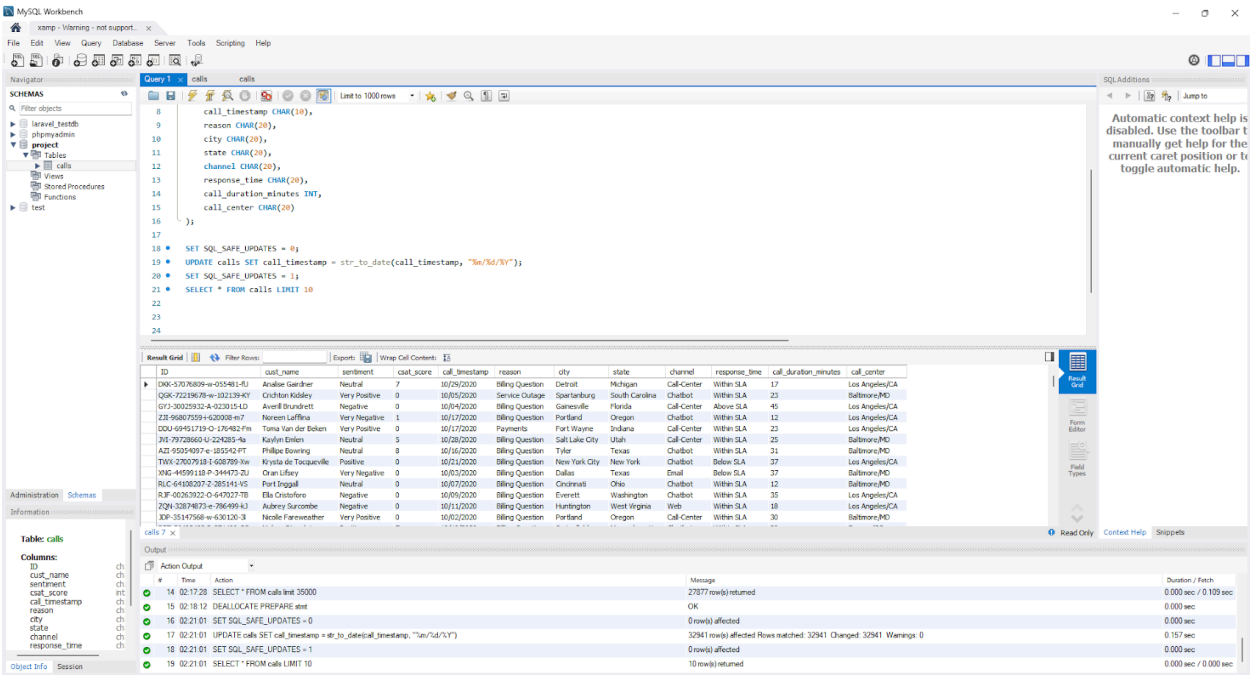
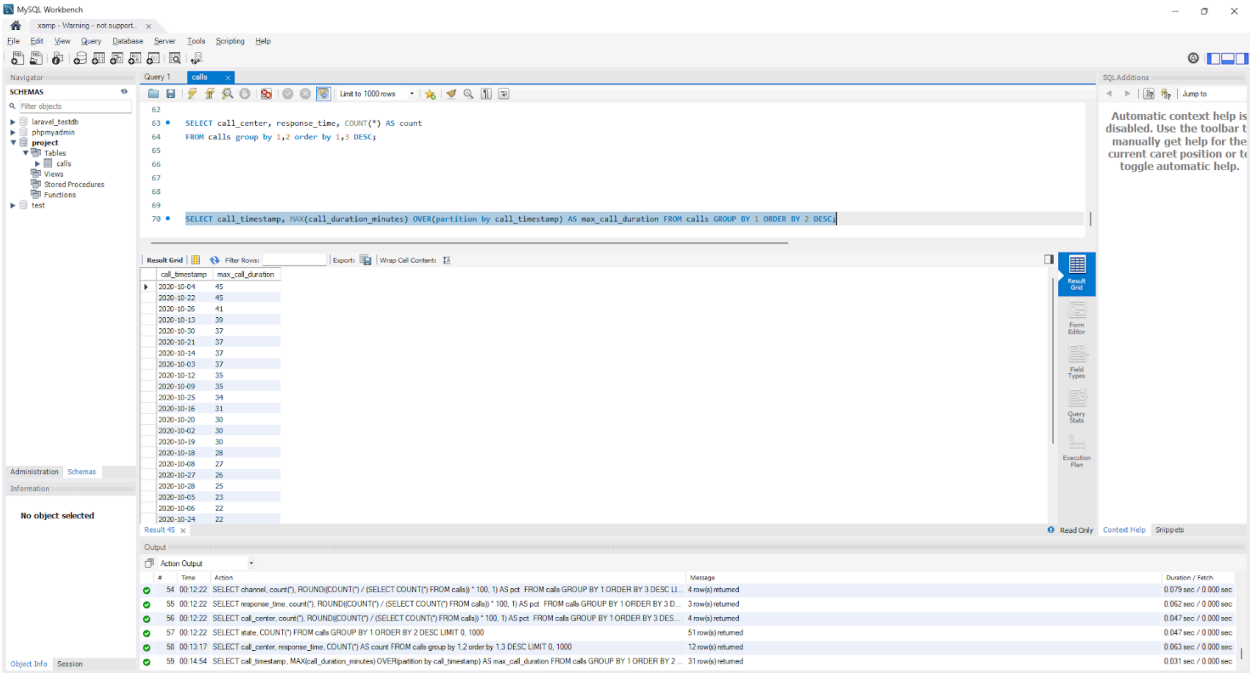
Finals Task 1. Data Preparations and Analysis using SQL

In this project, I built a SQL analytics script to clean and standardize raw call center timestamps, applied descriptive aggregation and window functions to evaluate peak durations and SLA compliance across locations, and broke down customer volume by day of the week to uncover key operational trends.

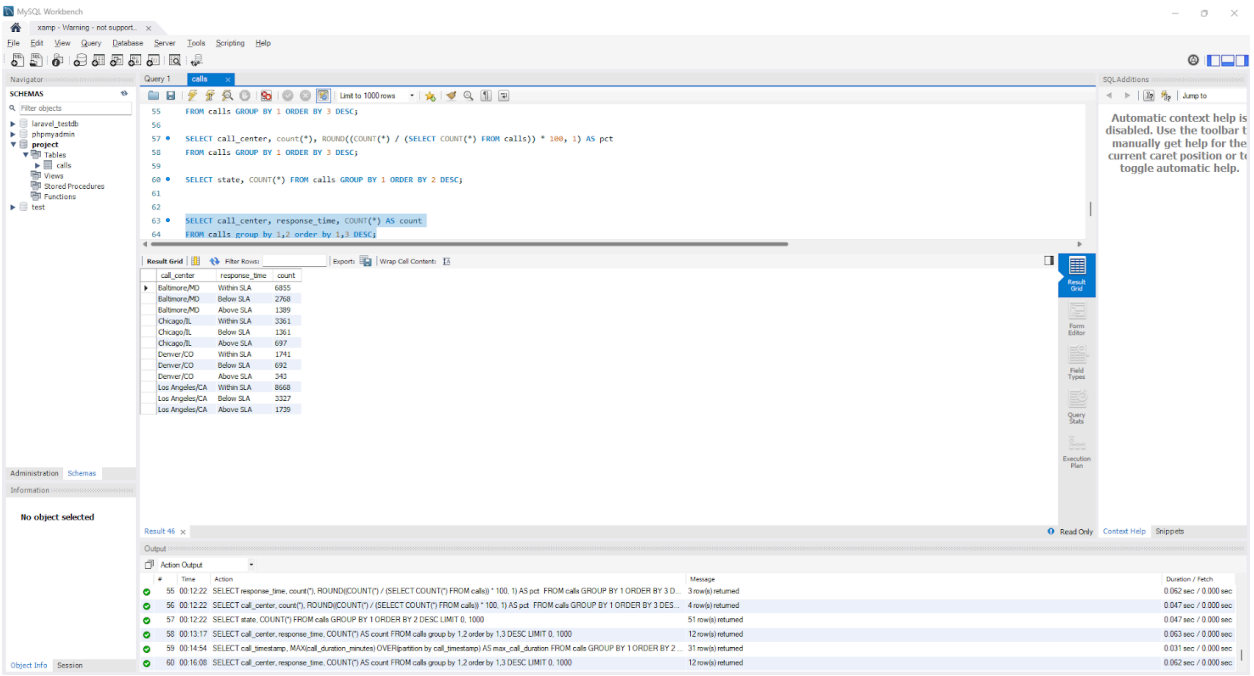
In this image, I performed the data cleaning phase by converting the text-based `call_timestamp` field into a standard SQL date format using the `str_to_date()` function after temporarily disabling safe update settings.



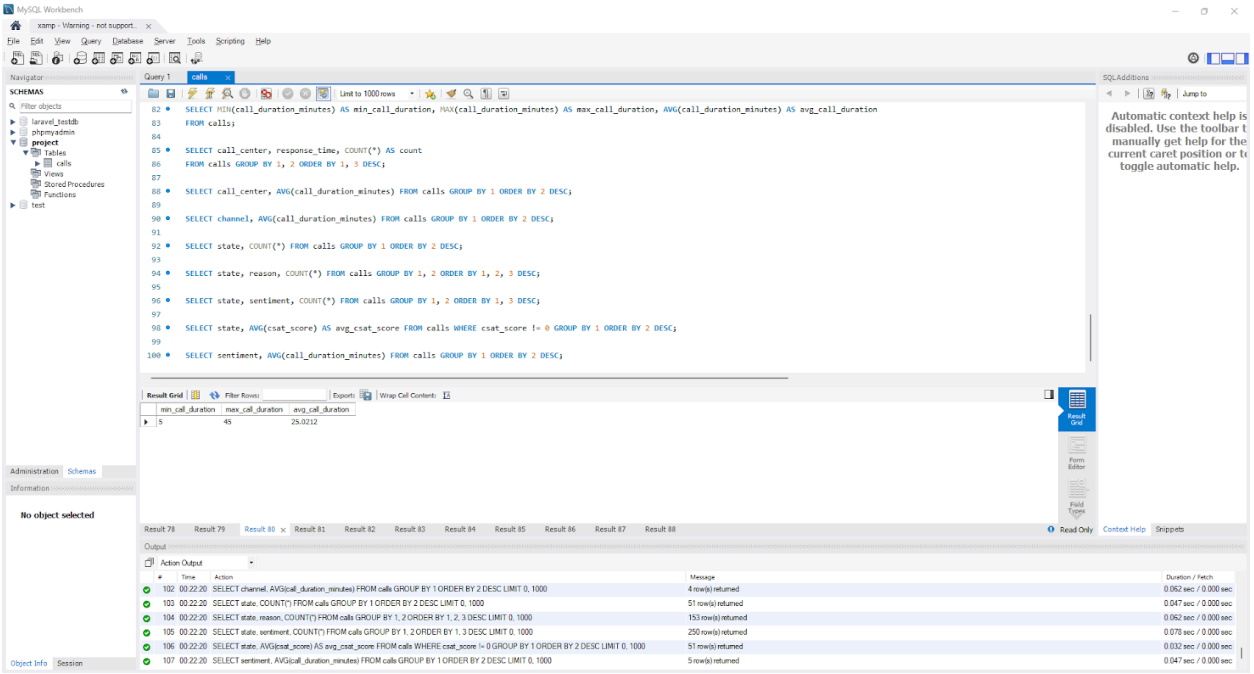
In this image, I tracked peak call lengths over time by applying the `MAX() OVER (PARTITION BY ...)` window function to calculate the maximum call duration for each unique timestamp.



In this image, I generated a breakdown of service level agreement compliance across different locations, counting how many calls fell within, below, or above SLAs for each individual call center.



In this image, I calculated high-level descriptive metrics for the dataset, revealing the minimum, maximum, and overall average call duration alongside a series of my exploratory diagnostic queries.



In this image, I extracted the day name from the timestamp column to calculate the total volume of calls handled per day of the week, which revealed to me that Friday was the busiest day.